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Wall hanging bracket set for supporting a weight training sled in an elevated vertical stowage position

Abstract

A wall hanging bracket set for supporting a weight training sled in an elevated vertical stowage position. The wall hanging bracket set includes an upper bracket for attachment to a vertical surface and supportively contacting a second longitudinal end of the sled at two laterally spaced contact points, and a lower bracket for attachment to the vertical surface and supportively contact a first longitudinal end of the sled at a single contact point.

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Background/Summary

BACKGROUND

(1) Weight sleds have become an increasingly popular exercise device in indoor health and fitness clubs, many of which have limited open space. Weight sleds, also known as blocking sleds, typically support weights upon one or more skids which exert frictional resistance against movement of the sled. An exemplary traditional weight sled is depicted in US Patent Application Publication 2014/0073492. Weight sleds were originally designed for outdoor use where space and the damage caused by the frictional sliding of the skids against the ground were of little concern. The transition from outdoor to indoor use has come with certain challenges, including the need for substantial open space and installation of flooring that can withstand the abrasive effects of repetitive frictional sliding of the skids over the flooring.

(2) Wheeled version of blocking sleds are depicted in U.S. Pat. No. 3,326,553 (a three wheeled skid-steer version) and U.S. Pat. No. 6,942,585 (a single wheel version) whereby frictional skidding is substantially eliminated, but at the expense of a loss in stability when pushing the sled—resulting in the need for an onboard operator to steer the sled of U.S. Pat. No. 3,326,553, or the need for additional space to accommodate the uncontrolled instability of the sled of U.S. Pat. No. 6,942,585.

(3) Stable wheeled weight training sleds are disclosed in U.S. Pat. Nos. 10,398,926, 10,398,927 and 10,799,750. While these weight training sleds constitute a significant advance over prior weight training sleds, a need still exists for a compact, lighter weight wheeled weight training sled.

SUMMARY OF THE INVENTION

(4) The invention is directed to a weight training sled and wall hanging bracket set for supporting the weight training sled in an elevated vertical stowage position.

(5) The weight training sled is a wheeled weight training sled that includes (a) a chassis having longitudinally spaced first and second ends and laterally spaced first and second sides, (b) at least two longitudinally spaced rotatable wheels for supporting the chassis a vertical distance above a support surface, (c) a brake for applying resistance to rotation of at least one of the wheels, (d) a first pair of laterally spaced push handles extending upward from proximate a first longitudinal end of the chassis, and (e) a second pair of laterally spaced push handles extending from proximate a second longitudinal end of the chassis wherein the first pair of push handles are pivotable as a unit about a laterally extending pivot axis as between an upward use position and a lowered stowage position extending towards the second longitudinal end of the chassis.

(6) The wall hanging bracket set, comprising (a) an upper laterally elongated bracket configured and arranged for attachment to a vertical surface and supportively contact a second longitudinal end of the sled at two laterally spaced contact points when attached to the vertical surface, and (b) a lower laterally elongated bracket configured and arranged for attachment to the vertical surface and supportively contact a first longitudinal end of the sled at a single contact point when attached to the vertical surface at a defined distance away from and vertically below the attached upper bracket.

(7) In a preferred embodiment, the wall hanging bracket set includes at least one hook or loop available for use in attaching one end of a battle rope or resistance band while a weight training sled is supported upon the hanging bracket set. The at least one hook or loop can be provided on either the upper or lower bracket, with a preference for providing at least one hook or loop on each of the upper and lower brackets.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front perspective view of one embodiment of a weight training sled in accordance with the invention, with the first pair of push handles pivoted into the use position.
- (2) FIG. 2 is a partially exploded perspective view of the invention depicted in FIG. 1 sans rear wheels and the second pair of push handles, with the first pair of push handles pivoted into the stowage position.
- (3) FIG. 3 is a partially exploded perspective view of the invention depicted in FIG. 1 sans second pair of push handles, with the first pair of push handles pivoted into the stowage position.
- (4) FIG. 4 is a perspective view of the invention depicted in FIG. 1 with detached second pair of push handles, with the first pair of push handles pivoted into the stowage position.
- (5) FIG. 4A is an enlarged view of a portion of the invention depicted in FIG. 4 encircled by A.
- (6) FIG. 5 is a rear perspective view of the invention depicted in FIG. 1 with the first pair of push handles pivoted into the use position.
- (7) FIG. 6 is a front perspective view of the invention depicted in FIG. 1 depicting the use and stowage pivot positions of the first pair of push handles.
- (8) FIG. 7 is a side view of the invention depicted in FIG. 1.
- (9) FIG. 8 is a perspective view of the invention depicted in FIG. 1 with a pull strap attached to the tow hook.
- (10) FIG. 8A is an enlarged view of a portion of the invention depicted in FIG. 8 encircled by A.
- (11) FIG. 9 is a perspective view of the invention depicted in FIG. 1 with a pull rope attached to the tow hook.
- (12) FIG. 9A is an enlarged view of a portion of the invention depicted in FIG. 9 encircled by A.
- (13) FIG. 10 is a front perspective view of one embodiment of a wall hanging bracket set in accordance with the invention.
- (14) FIG. 11 is a front view of the invention depicted in FIG. 10.
- (15) FIG. 12 is a front perspective view of the weight training sled depicted in FIG. 3 and the wall hanging bracket set depicted in FIG. 10 in hanging spatial relationship to one another.
- (16) FIG. 13 is a front perspective view of the weight training sled depicted in FIG. 3 hanging from the wall hanging bracket set depicted in FIG. 10 and detached second push handles shown in mounted spatial relationship to the wall hanging bracket set.
- (17) FIG. 14 is an enlarged front perspective view of the invention depicted in FIG. 10.
- (18) FIG. 15 is a perspective view of the invention depicted in FIG. 1 with the first pair of push handles pivoted into the stowage position.
- (19) FIG. 15A is an enlarged view of a portion of the invention depicted in FIG. 15 encircled by A.
- (20) FIG. 16 is a bottom view of the invention depicted in FIG. 1 with the first pair of push handles pivoted into the stowage position.
- (21) FIG. 17 is a bottom perspective view of the invention depicted in FIG. 16.

DETAILED DESCRIPTION OF A PREFERRED EMBODIMENT

Definitions

- (22) As utilized herein, including the claims, the term “neutral resistance” means resistance at or near zero, whereby the wheeled exercise sled of the present invention is rendered suitable for use as a wheeled transport wagon when the braking mechanism is set to neutral.
- (23) TABLE-US-00001 Nomenclature Table
REF. NO. NAME
100 Wheeled Weight Training Sled
101 First End of Weight Training Sled
102 Second End of Weight Training Sled
103 First Side of Weight Training Sled
104 Second Side of Weight Training Sled
110 Chassis
111 First End of Chassis
112 Second End of Chassis
113 First Side of Chassis
114 Second Side of Chassis
115 Top of Chassis
116 Bottom of Chassis
117 First Side Rail of Chassis
117a First End of First Side Rail
117b Second End of First Side Rail
118 Second Side Rail of Chassis
118a First End of Second Side Rail
118b Second End of Second Side Rail
119 Cross Beams Interconnecting Side Rails
120 Axles

121 First Axle 122 Second Axle 130 Wheels 131 Front Wheel(s) 132 Rear Wheel(s) 140 Push Handles 140d Distal End of Push Handles 140p Proximal End of Push Handles 141 First Pair of Push Handles 142 Second Pair of Push Handles 143 Grips on Each Push Handle 145 Interconnecting Member 146 Pivot Axis for First Pair of Push Handles 147 Locking Pin for First Pair of Push Handles 149 Locking Pin on Each Second Push Handle 150 Weight Plate Horn 160 Tow Hook 170 Posts or Sleeves for Second Push Handles 170z Central Axis of Post or Sleeve 179 Locking Slot in Each Post or Sleeve 180 Braking Mechanism 185 Braking Level Adjustment Lever 190 Rubber Bumpers 200 Wall Hanging Bracket Set 201 Upper Bracket 201h Upper Bracket Hook or Loop 201p Upper Bracket Contact Support Points 202 Lower Bracket 202h Lower Bracket Hook or Loop 202p Lower Bracket Contact Support Point 203 Posts or Sleeves for Second Push Handles 300 Elongate Flexible Exercise Pull (e.g. pull strap, battle rope, elastic band, etc.) x Longitudinal Direction y Lateral Direction z Transverse Direction
Construction

(24) With reference to the illustrative drawings, the invention is directed to a wheeled weight training sled **100** (hereinafter “sled”) and a wall hanging bracket set **200** for supporting a weight training sled such as sled **100** in an elevated vertical stowage position.

(25) Wheeled Sled

(26) Referring to FIGS. **1-9** and **15-17**, the wheeled sled **100** includes a chassis **110**, at least two wheels **130**, two pair of push handles **140** with a pair proximate each longitudinal end **101** and **102** of the sled **100**, and at least one braking mechanism **180**. The wheeled sled **100** preferably includes at least one of (i) three fixed-directional wheels **130** comprised of a single front wheel **131** mounted upon a first axle **121** proximate the first longitudinal end **101** of the sled **100** and a pair of rear wheels **132** mounted upon a second axle **122** proximate the second longitudinal end **102** of the sled **100**, (ii) a push handle pivot feature wherein a first pair of the push handles **141** proximate the front or first longitudinal end **101** of the sled **100** is pivotable as a unit about a laterally y extending pivot axis **146** as between an upward use position and a lowered stowage position extending towards the rear or second longitudinal end **102** of the sled **100**, (iii) a tow hook **160** mounted between and pivotable with the first pair of push handles **141**, and (iv) a push handle conformation adjustment feature wherein the second pair of push handles **142** are curvilinear and releasably engageable with a post or sleeve **170** extending upward from the chassis **110**, and rotatable up to 180° about a central axis **170z** of each post or sleeve **170** as between a first axially rotated locked position and a second axially rotated locked position.

(27) The wheeled sled **100** has longitudinally x spaced first and second ends **101** and **102**, and laterally y spaced first and second sides **103** and **104**.

(28) Referring to FIGS. **2, 3** and **16**, a preferred chassis **110** is a metal structure having first and second longitudinally x elongated and laterally y spaced side rails **117** and **118** rigidly interconnected by cross-beams **119**, defining a chassis **110** with first and second longitudinal ends **111** and **112**, first and second lateral sides **113** and **114**, and a transversely z spaced top **115** and bottom **116**.

(29) Referring to FIGS. **1, 3-9** and **15-17**, the wheels **130** are rotatably mounted to the chassis **110** for supporting the bottom **116** of the chassis **110** a distance above a surface (hereinafter referenced as “clearance”). The wheels **130** can be fixed-directional wheels **130** so as to constrain the chassis **110** to reciprocating travel upon a surface along a substantially linear longitudinal x path.

(30) When two wheels **130** are employed they are preferably longitudinally x aligned in the midsagittal plane of the sled **100**. When three wheels **130** are employed they are preferably spaced at the corners of an isosceles triangle with two of the wheels **130** laterally y aligned proximate one end **102** of the sled **100** and the third centrally positioned proximate the other end **101** of the sled **100**. When four wheels **130** are employed the wheels **130** are mounted in laterally y spaced pairs upon each of two laterally y extending axles **121** and **122** mounted proximate each longitudinal end **101** and **102** of the sled **100** with the wheels **130** in each pair of wheels mounted proximate

opposite sides **103** and **104** of the sled **100**. Alternatively, the four wheel embodiment may employ a pair of longitudinally x aligned and laterally y centered wheels **130** proximate the longitudinal ends **101** and **102** of the sled **100**, with a vertically z raised or vertically z aligned outrigger wheel **130** extending from each side **103** and **104** of the sled **100**.

(31) The wheels **130** are preferably pneumatic wheels **130** with good traction in order to limit undesired sliding of the wheels **130** across the floor during exercise as opposed to desired rotation of the wheels **130**.

(32) Referring to FIGS. **1**, **4-9** and **15-17**, the sled **100** includes two pair of push handles **141** and **142**, with a first pair of push handles **141** secured to the first end **101** of the sled **100**, preferably secured to the first ends **117a** and **118a** of the chassis side rails **117** and **118**, and a second pair of push handles **142** secured to the second end **102** of the sled **100**, preferably secured to the second ends **117b** and **118b** of the chassis side rails **117** and **118**. This allows a user to exercise by pushing the sled in either direction along the longitudinal x path of travel.

(33) Referring to FIG. **6**, the first pair of the push handles **141** can be configured and arranged to pivot as a unit about a laterally y extending pivot axis **146** as between an upward use position and a lowered stowage position extending towards the rear or second longitudinal end **102** of the sled **100**. The first pair of push handles **141** can be releasably locked in the upward use position and a lowered stowage position by a spring biased locking pin **147** attached to the first pair of push handles **141** for pivoting with the first pair of push handles **141**, which can be selectively inserted into at least two radially spaced orifices (not numbered) attached to the chassis **110**.

(34) Referring to FIGS. **4** and **4A**, the second pair of push handles **142** can be configured and arranged to allow conformational adjustment of the push handles **142** on the sled **100** by releasably engaging each of the push handles **142** to the sled **100** via a post or sleeve **170** extending upward from the chassis **110**, and configuring the push handles **142** and the posts or sleeves **170** with cross-sectional shapes that allows each push handle **142** to be engaged with a post or sleeve **170** at different rotated angles of up to 180° about a central axis **170z** of each post or sleeve **170** thereby allowing mounting of the push handles **142** as between a first axially rotated locked position and a second axially rotated locked position. The second pair of push handles **142** can be releasably locked in each axially rotated position by a locking pin **149** at the proximal end **140p** of each handle **142** that engages locking slots **179** in the distal end (unnumbered) of each post or sleeve **170**.

(35) Referring to FIGS. **1**, **4-9** and **15-17**, the second pair of push handles **142** are preferably curvilinear, with grips **143** proximate the distal ends **140d** of the push handles **142** that angle inward towards one another and downward towards the chassis **110** when in one of the axially rotated locked positions.

(36) Inward angling of the grips **143** provides a more natural ergonomic rotational gripping position, while downward angling of the grips **143** redirects at least some of the vertical force vector created when a user is pushing the sled **100** from an upwardly directed force vector to a downwardly directed force vector, thereby preventing or at least limiting lifting of the work end of the sled **100** off the ground.

(37) Referring to FIGS. **8**, **8A**, **9** and **9A**, a tow hook **160** can be mounted between and pivotable with the first pair of push handles **141**. The distal end **140d** of the first pair of push handles **141** can be interconnected by an interconnecting member **145**. This interconnecting member **145** provides both stabilization of the paired push handles **141** and an elevated position for attachment of an elongate flexible exercise pull **300** such as a pull strap, battle rope, elastic band, etc. to the sled **100**. A tow hook **160** preferably extends longitudinally x outward from the lateral y center of this interconnecting member **145** to facilitate temporary attachment of an elongate flexible exercise pull **300**. The interconnecting member **145** can also be gripped by a user pushing the sled **100** as an alternative gripping position.

(38) Referring to FIGS. **8**, **8A**, **9** and **9A**, the interconnecting member **145**, and thereby the tow

hook **160**, is preferably located so as to provide a clearance of at least 30 cm between the tow hook **160** and ground when the first pair of push handles **141** are pivoted into the use position. Such elevated positioning of the tow hook **160** serves to limit the amount of upward force vector created when a user pulls upon a elongate flexible exercise pull **300** attached to the sled **100** at the tow hook **160**, thereby limiting and potentially eliminating lifting of the first or front end **101** of the sled **100** off the ground.

(39) Referring to FIGS. **1-9** and **15**, one or more weight plate horns **150** can be provided on the chassis **110** for mounting weight plates (not shown) onto the top **115** of the chassis **110** in order to increase exercise resistance offered by the sled **100** and, more importantly, counteract any upward lifting force vector exerted by a user that would tend to lift an end of the sled **100** and thereby lift the wheel(s) **130** closest to the user off the floor. The preferred embodiment has a weight plate horn **150** positioned along the midsagittal plane of the sled **100** secured to a cross beam **119**. Rubber pads (not shown) can be provided atop the chassis **115** proximate the horn **150** for cushioning and protecting the chassis **110** when weight plates are added to or removed from the horn **150**.

(40) Referring to FIGS. **1-9** and **15-17**, a braking mechanism **180** is attached to the chassis **110** and in communication with at least one of the wheels **130**, preferably in communication with the first or front wheel **131**, for exerting a bidirectional controlled variable resistive force against rotation of the wheel(s) **130** along the longitudinal x path of travel. Separate braking mechanisms **201** and **202** can be provided for each wheel **130** or each axle **120**. Many types of resistance devices are known such as braking motors, generators, brushless generators, eddy current systems, magnetic systems, alternators, tightenable belts, friction rollers, fluid brakes, etc., any of which could be effectively utilized in the present invention. A braking mechanism capable of providing progressive resistance based upon acceleration or speed of travel is generally preferred.

(41) In further detail, a preferred braking mechanism **180** is an eddy current brake mounted to a side rail **117** or **118** of the chassis **110** for exerting resistance to rotation of the first axle **121**. An eddy current brake generally includes (i) a mounting plate which would be rigidly attached to the chassis **110**, (ii) a drive sprocket rotatably with and secured to a portion of the first axle **121** extending through an orifice in the mounting plate, (iii) a pulley assembly with a pulley and driven sprocket rotatably mounted via an internal hub and a sprocket mount onto a first mounting post projecting from the mounting plate in rotatable driven communication with the drive sprocket via a drive chain, (iv) an eddy disk assembly comprised of an eddy disk rotatably mounted via a disk mount having a shaft and mounting plate onto a second mounting post projecting from the mounting plate in rotatable driven communication with the pulley assembly via a drive belt, (v) a drive belt tensioning assembly secured within an adjustment slot in the mounting plate for adjustably tensioning the drive belt, and (vi) a magnetic stator assembly secured to the mounting plate for manual (as shown) or automatic (not shown) repositioning of the magnets relative to the eddy disk of the eddy disk assembly via an adjustment slot in the mounting plate or a multi-stop lever **185** as depicted in FIGS. **15** and **15A**, to increase or decrease resistance as desired.

(42) Referring to FIGS. **15** and **15A**, in a preferred embodiment the braking mechanism **180** is adjustable into a neutral resistance setting, whereby the sled **100** is effectively converted from an exercise sled to a transport wagon. The neutral setting facilitates movement of the sled **100** from one location to another, such as transport back and forth between a storage location and a use location. The neutral setting preferably applies some modest resistance to rotation of the wheels which does not appreciably interfere with transport of the sled **100** but is effective for preventing or at least slowing down gravity induced movement of the sled **100**.

(43) A protective shroud (unnumbered) may be provided over the components of the braking mechanism **180**.

(44) Rubber bumpers **190** can be provided on the front end **101** of the sled **100** to minimize any structural damage should a user strike an object while pushing the sled **100**.

(45) Various acceptable, preferred and most preferred dimensions having some significance to the

value and/or performance of the sled **100** are provided below.

(46) TABLE-US-00002 Acceptable Preferred Most Preferred Dimension (cm) (cm) (cm) Chassis Longitudinal 60-150 >100 100-140 Length of Chassis Maximum Lateral 30-100 40-80 50-60 Width of Chassis Maximum Transverse 3-30 5-20 10-20 Height of Chassis Clearance >2 5-20 10-15 Wheels Wheelbase 60-100 >80 80-100 Track 40-100 60-100 80-100 Second Push Handles Height relative to Floor 40-120 50-100 60-100 Lateral Spacing 70-120% 80-100% 80-90% at Top End of Track of Track of Track

Wall Hanging Bracket Set

(47) Referring to FIGS. **10-14**, the wall hanging bracket set **200** includes an upper laterally y elongated bracket **201** and a lower laterally y elongated bracket **202**. The upper bracket **201** is configured and arranged for attachment to a vertical surface and for supportively contacting the second longitudinal end **102** of the sled **100** at two laterally y spaced contact points **201p** when attached to the vertical surface. The contact points **201p** on the upper bracket **201** preferably engage the chassis **110** of the sled **100**. The lower bracket **202** is configured and arranged for attachment to the same vertical surface and, when attached to the vertical surface at a defined distance away from and vertically below the attached upper bracket **201**, can supportively contact a first longitudinal end **101** of the sled **100** at a single contact point **202p**. The contact point **202p** on the lower bracket **202** preferably engages the front wheel **131** of the sled **100**.

(48) Referring to FIGS. **10-14**, the lower bracket **202** preferably includes upward projecting sleeves or posts **203** upon which the second pair of push handles **142** can be secured for storage when detached from the chassis **110**.

(49) Referring to FIGS. **10** and **12-14**, the wall hanging bracket set **200** preferably has a low mounted horizontal projecting profile, with each of the upper **201** and lower **202** brackets extending less than 6 inches away from a vertical surface upon which the brackets **201** and **202** are attached.

(50) In a preferred embodiment, the wall hanging bracket set **200** includes at least one hook or loop **201h** and/or **202h** available for use in attaching one end of an elongate flexible exercise pull **300** while a weight training sled **100** is supported upon the hanging bracket set **200**. The at least one hook or loop **201h** and/or **202h** can be provided on either the upper **201** or lower **202** bracket, with a preference for providing at least one hook or loop **201h** and **202h** on each of the upper **201** and lower **202** brackets respectively.

(51) Use

(52) The sled **100** can be conveniently and safely used in a confined space as small as 1.2 meters wide and 5 meters long, by (i) setting the braking mechanism(s) **180** to the desired resistance, (ii) standing at the second end **102** of the sled **100**, (iii) leaning forward and gripping the second pair of push handles **142**, (iv) pushing the sled **100** in a first longitudinal x direction along a longitudinal path, (v) walking around the sled **100** to the first end **101** of the sled **100**, (vi) leaning forward and gripping the first pair of push handles **141**, (vii) pushing the sled **100** in a second longitudinal x direction back along the longitudinal path, (viii) walking back around the sled **100** to the second end **102** of the sled **100**, and (ix) repeating steps (iii)-(viii) for as many reps as desired.

(53) Alternatively, the sled **100** can be conveniently and safely used in a confined space as small as 1.2 meters wide and 5 meters long, by (i) setting the braking mechanism(s) **180** to the desired resistance, (ii) standing at the second end **102** of the sled **100**, (iii) leaning forward and gripping the second pair of push handles **142**, (iv) pushing the sled **100** in a first longitudinal x direction along a longitudinal path, (v) tilting the first end **101** of the sled **100** up until the front wheel **131** loses contact with ground, (vi) rotating the sled **101** 180° with only the rear wheels **132** in contact with ground, (vii) pushing the sled **100** in a second longitudinal x direction back along the longitudinal path, and (viii) repeating steps (v)-(vii) for as many reps as desired.

Claims

1. A wall hanging bracket set for supporting a weight training sled having a chassis and a pair of push handles detachable from the chassis in an elevated vertical stowage position, comprising (a) an upper laterally elongated bracket configured and arranged for attachment to a vertical surface and supportively contact a second longitudinal end of the sled at two laterally spaced contact points when attached to the vertical surface, and (b) a lower laterally elongated bracket configured and arranged for attachment to the vertical surface and supportively contact a first longitudinal end of the sled at a single contact point when attached to the vertical surface at a defined distance away from and vertically below the attached upper bracket, the lower laterally elongated bracket further including a pair of laterally spaced and upward projecting sleeves or posts operable for securing the pair of push handles when detached from the chassis in an upwardly projecting storage orientation.
 2. The wall hanging bracket set of claim 1 wherein the upper bracket further comprises at least one hook or loop operable for use in attaching one end of a battle rope or resistance band.
 3. The wall hanging bracket set of claim 2 wherein the at least one hook or loop is available for use in attaching one end of a battle rope or resistance band with a weight training sled supported upon the upper bracket.
 4. The wall hanging bracket set of claim 1 wherein the lower bracket further comprises at least one hook or loop operable for use in attaching one end of a battle rope or resistance band.
 5. The wall hanging bracket set of claim 4 wherein the at least one hook or loop is available for use in attaching one end of a battle rope or resistance band with a weight training sled supported upon the lower bracket.
 6. The wall hanging bracket set of claim 1 wherein each of the upper and lower brackets extend less than 6 inches away from a vertical surface upon which the bracket is attached.
 7. The wall hanging bracket set of claim 1 wherein the upper laterally elongated bracket is configured and arranged to supportively contact a chassis of the sled.
 8. The wall hanging bracket set of claim 7 wherein the lower laterally elongated bracket is configured and arranged to supportively contact a wheel on the sled.
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